

Land Surface Reflectance

G2GS

Exported on May 14, 2021

Table of Contents

1	Introduction	4
1.1	Purpose of This Document	4
1.2	Who Should Use This Document	4
1.3	Inside Each Section	4
1.4	Related Documents	4
1.5	Revision History	5
1.6	List of Acronyms	5
2	Algorithm Description	6
2.1	Algorithm Overview	6
2.1.1	Product Generated	6
2.1.2	GOCI-II Characteristics	7
2.2	Processing Outline	8
2.3	Algorithm Input	8
2.3.1	Primary Sensor Data	9
2.3.2	Ancillary Data	9
2.3.3	Derived Data	9
2.4	Theoretical Description	9
2.5	Mathematical Description	10
2.5.1	Structure and Interpolation of LUT data	10
2.5.2	GOCI-II Surface Reflectance (G2SR)	11
2.6	Algorithm Product Output	12
2.6.1	Output	12
2.6.2	Flags	12
2.6.3	Metadata	13
3	Test Data Sets And Output	14
3.1	Simulated / Proxy Input Datasets	14
3.2	Output from Simulated Datasets	14
3.2.1	Accuracy Estimates	14
3.2.2	Error Budget	15
4	PRACTICAL CONSIDERATIONS	16
4.1	Numerical Computation Considerations	16
4.2	Programming and Procedural Considerations	16
4.3	Quality Assessment and Diagnostics	16
4.4	Exception Handling	16
4.5	Algorithm Validations	16
5	ASSUMPTIONS AND LIMITATIONS	17
5.1	Performance	17
5.2	Assumed Sensor Performance	17
6	Pre-Planned Product Improvement	18
7	REFERENCES	19
8	APPENDIX	20

Document Title	Surface Reflectance Algorithm for GOCI-II Data
Initial creation date	2017-11-20
Last update date	2021-05-07
Writer	Kwon-Ho Lee ¹ and Kyeong-Sang Lee ² ¹ Gangneung-Wonju National University, Gangneung, Republic of Korea ² Korea Institute of Ocean Science and Technology, Busan, Republic of Korea

1 Introduction

1.1 Purpose of This Document

The present Algorithm Theoretical Basis Document (ATBD) is to describe the surface reflectance retrieval algorithm using the Geostationary Ocean Color Imager II (GOCI-II) sensor on board the GK-2B satellite, which will be launched in 2019. The surface reflectance calculation algorithm described in this document can work when the qualified data are available after the successful launch of GK-2B satellite. Thus, the algorithm uses proxy data (e.g. currently operated GOCI data) as input data of the algorithm. When data of GOCI-II Level1B (L1B) are sufficiently acquired after the launch, the present algorithm will retrieve and redistribute surface reflectance products by the GOCI-II Ground Segment system (G2GS) Algorithm Development Environment (ADE).

1.2 Who Should Use This Document

The intended users of this document are those who interested in understanding the physical basis of the algorithm and use the surface reflectance retrieval outputs of this algorithm. This document also provides information useful to anyone maintaining or modifying the original algorithm.

1.3 Inside Each Section

This document is broken down into the following main sections:

- **Introduction:** Provides the objectives of the GOCI-II surface reflectance retrieval algorithm, relevant characteristics of the instruments, and provides a brief description of the products generated by this algorithm.
- **Algorithm Description:** Provides detailed description of the physical and mathematical basis of the algorithm, the input and output data.
- **Test Datasets and Outputs:** Describe the test data sets and processes to characterize the performance of the algorithm. This includes the method used for proxy GOCI data, algorithm output from the proxy GOCI data.
- **Practical Considerations:** Provides the considerations for the S/W maintenances for the algorithm involving numerical computation, programming and procedures, quality assessment and diagnostics, and exception handling.
- **Assumptions and Limitations:** Provides the current limitations of the approach and gives the plan for overcoming these limitations with further algorithm development.

1.4 Related Documents

Besides the references given throughout, this document is related to documents listed as below:

- GOCI-II ATBD for Vicarious Calibration

1.5 Revision History

This is the updated second version (Version 0.4) of this document after 100% maturity delivery. Version 0.4 is based on Version 0.3 including only the revisions and correction of potential errors.

Revision	Date	Revision Description	Page Affected
0.2	2017-11-20	Initial Issue	
0.3	2018-04-30	Minor changes in methodology	
0.4	2018-11-28	Minor changes in methodology	
0.5	2021-05-07	Correction of some typing errors	all

1.6 List of Acronyms

Table 1. List of acronyms

GOCI-I	Geostationary Ocean Color Imager-I
GOCI-II	Geostationary Ocean Color Imager-II
GK-2B	Geostationary KOMPSAT 2B
UV	Ultra Violet
TOMS	Total Ozone Mapping Spectrometer
GOME	Global Ozone Monitoring Experiment
MODIS	MODerate resolution Imaging Spectroradiometer
OMI	Ozone Monitoring Instrument
LER	Lambert Equivalent Reflectivity
LUT	Look-Up Table
RMSE	Root Mean Square Error
ACR	Atmospherically Corrected Reflectance
SRP	Surface Reflectance Product
G2GS	GOCI-II Ground Segment system
ADE	Algorithm Development Environment
SNR	Signal-to-Noise Ratio

2 Algorithm Description

2.1 Algorithm Overview

Surface reflectances are the most widely used common products in the satellite's operation products. For example, well-known earth observing satellites such as Total Ozone Mapping Spectrometer (TOMS), Global Ozone Monitoring Experiment (GOME), MODerate resolution Imaging Spectroradiometer (MODIS), and Ozone Monitoring Instrument (OMI) have been widely used. Among them the TOMS has the earliest observation records from the first TOMS on board the Nimbus-7 satellite launch in 1979. TOMS can calculate the Lambert Equivalent Reflectivity (LER) under the certain atmospheric conditions (i.e. pristine atmosphere). The LER value at 340nm and 380nm are less dependent on the wavelength (<0.2%) and a range of LER values at these wavelengths are known to be relatively low (land: 2-4%, sea surface: 5-6%) (Herman and Colarior, 1997). However, the regions without clouds or aerosols are rarely observed due to the relatively large horizontal spatial resolution (~50km at nadir; ~250km off nadir) of the TOMS sensor so that LER is hard to be derived persistently.

Since 1995, the GOME of the European Remote Sensing-2 (ERS-2) satellite has been started produce LER product. GOME sensor is a spectrometer having spectral range of 237-772 nm with a spatial resolution of 40 km 320 km. The GOME algorithm calculates the LER at 11 wavelengths (335nm, 380nm, 416nm, 440nm, 463nm, 494.5nm, 555nm, 610nm, 670nm, 758nm, and 772nm) selected among the spectral observation data (Koelemeijer et al., 2003).

The OMI on board the Aura satellite has been operated from October 2004, and it observes a continuous wavelength range of 270-500nm at 13km 24km spatial resolution. The surface reflectance values on a basis of three-year accumulated OMI data (October 2004 to October 2007) has been calculated and used for the other retrieval algorithm (<http://mirador.gsfc.nasa.gov/>). Compared OMI LER to the surface reflectance retrieved from other satellite, the OMI LER data was differed as 1.5% and 0.5% for the surface reflectance from TOMS and GOME, respectively (Kleipool et al., 2008).

The MODIS sensors from Terra (2000 – current) and Aqua (2002 – current) satellites have been used to retrieve surface reflectances. MODIS algorithm uses atmospheric Rayleigh correction, which were inputted to radiative transfer models (RTMs). Surface reflectance values at 0.47 μ m, 0.55 μ m, 0.66 μ m, 0.85 μ m, 1.2 μ m, 1.6 μ m, and 2.1 μ m were calculated in the output products of MODIS level 2 (L2) surface reflectance (Vermote et al., 2002). The MOD09/MYD09 data include the product that can predict the anisotropic reflection effect in the ground using bidirectional reflectance distribution function (BRDF) (Schaaf et al., 2002).

2.1.1 Product Generated

This section will describe the surface reflectance products generated by the GOCI-II surface reflectance algorithm and the requirements it places on the sensor.

1) Overview of the products

In general, surface reflectance is defined as of the fraction of incoming solar radiation and reflected radiance by Earth's surface. It's magnitude depends on the sun-satellite geometry, optical pathlength, surface materials, shape and so on. Surface reflectance is one of the important parameters that are closely related to weather, environments, and climate changes (Dickinson, 1983) as well as affecting the Earth's radiation budget (Chin et al., 2009). It is also used as input data to derive the variables in the geophysics, chemistry, and biology through the remote sensing systems (Veefkind et al., 2006; Noguchi et al., 2014). Therefore, derivation of the accurate surface reflectance from satellite observation data is required to develop algorithms for the satellite remote sensing and its applied studies.

2) Product requirements (spatial/temporal resolution, accuracy)

Requirements for the surface reflectance products are following.

- Consideration of the GOCI-II geometry

- If temporal composite of time series data is required, appropriate composite period considering GOCI-II observation time interval.
- required processing time for the near real time data distribute: within 10 minutes, G2GS ADE requirement

As described in Tables 2, accuracy is defined as 72% of the other operational products.

Table 2. GOCI II mission requirements for surface reflectance

Input	Output	Validation data	Accuracy (%)	cause
GOCI-1 L1B data	L2 SRP	MODIS L2 surface reflectance products	>72%	About 85% of the known accuracy reported from the operational products, consideration of the average aerosol optical thickness(AOT) over GOCI-II looking area (0.4 ~ 0.6)

2.1.2 GOCI-II Characteristics

Table 3 lists central wavelengths and bandwidths for the GOCI-II visible and near-infrared bands as well as designed Signal-to-Noise Ratio (SNR). It has 1 UV wavelength (380 nm), 8 visible wavelengths (412, 443, 490, 510, 555, 620, 660, 680 nm), 3 near infrared (NIR) wavelengths (709, 745, 865 nm), and 1 broadband (400-900nm) wavelength.

Table 3. Detailed characteristics of the GOCI-II

Central wavelength	Bandwidth	SNR @ specified input spectral radiance
380 nm	20 nm	998 @ 93.0 W m ⁻² μm ⁻¹ sr ⁻¹
412 nm	20 nm	1050 @ 100 W m ⁻² μm ⁻¹ sr ⁻¹
443 nm	20 nm	1145 @ 92.5 W m ⁻² μm ⁻¹ sr ⁻¹
490 nm	20 nm	1228 @ 72.2 W m ⁻² μm ⁻¹ sr ⁻¹
510 nm	20 nm	1180 @ 64.9 W m ⁻² μm ⁻¹ sr ⁻¹
555 nm	20 nm	1124 @ 55.3 W m ⁻² μm ⁻¹ sr ⁻¹
620 nm	20 nm	1102 @ 53.3 W m ⁻² μm ⁻¹ sr ⁻¹
660 nm	20 nm	1060 @ 32.0 W m ⁻² μm ⁻¹ sr ⁻¹
680 nm	10 nm	914 @ 27.1 W m ⁻² μm ⁻¹ sr ⁻¹
709 nm	10 nm	914 @ 27.7 W m ⁻² μm ⁻¹ sr ⁻¹
745 nm	20 nm	903 @ 17.7 W m ⁻² μm ⁻¹ sr ⁻¹
865 nm	40 nm	788 @ 12.0 W m ⁻² μm ⁻¹ sr ⁻¹
wideband	N/A	N/A

Basically, clear sky detection from the GOCI-II observations is required to retrieve surface reflectance. This can be attained by finding the smallest reflectance value during the given observation period. GOCI-II UV wavelength channel can be used to find UV-absorbing aerosols which can be helpful to atmospheric correction. Because the former ocean color sensors (i.e.

GOCI, SeaWiFS, etc.) cannot detect UV-absorbing aerosol, there is limitation over the area showing low atmospheric transmittance.

2.2 Processing Outline

Figure 1 shows the flow of GOCI-II surface reflectance retrieval algorithm using the minimum reflectance technique. The minimum reflectance technique calculates reflectance values for the clearest sky by finding the lowest reflectance with the atmospherically corrected reflectance series during a given period. It assumes that the surface reflection is uniform during the period. The effect of atmosphere, mixing gases, and aerosols is eliminated (or minimized) in this technique.

Operational GOCI-II algorithm is designed to be two parts: pre-process for the Atmospherically Corrected Reflectance (ACR) and near-real-time (NRT) process for the surface reflectance product (SRP). The L1B reflectance data is used as input data to detect clear sky pixel by the traditional cloud screening tests. The atmospheric correction can be performed with clear sky pixels (without cloud contaminated pixels) using the LUT. In the NRT process, the above ACR results are used as input data to accumulate data for a pixel and then the minimum values are found.

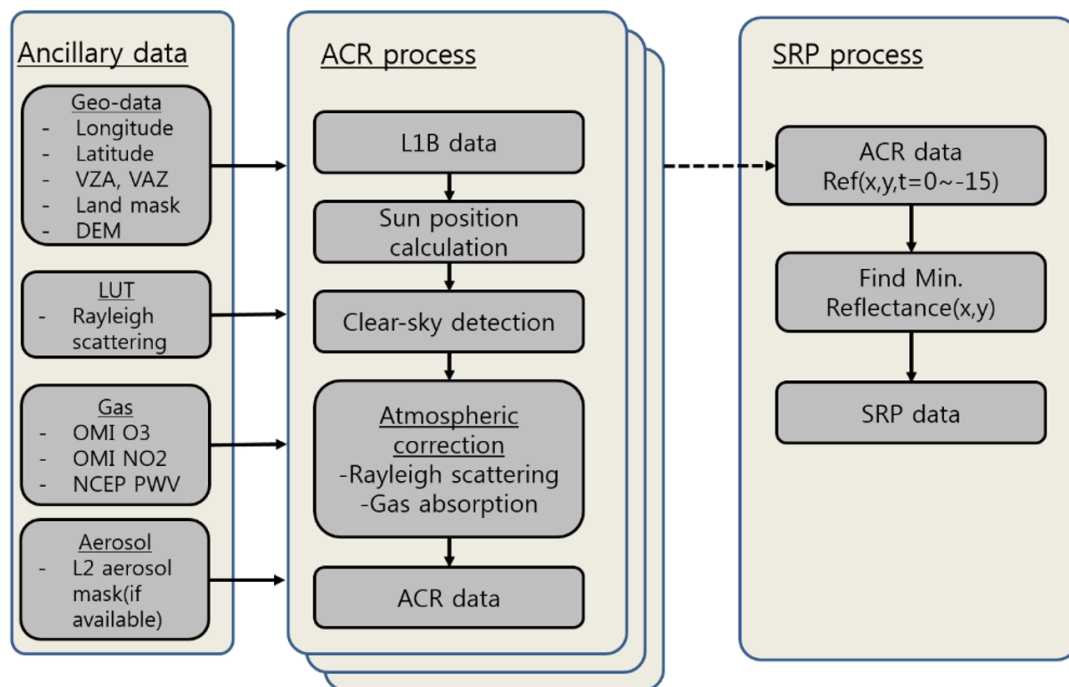


Figure 1. GOCI-II surface reflectance retrieval process.

2.3 Algorithm Input

The GOCI-II surface reflectance employs the calibrated L1b reflectance, geometry data, LUT, and trace gases (O₃, NO₂, H₂O). LUT is the satellite receiving radiance as a function of atmospheric condition, which were theoretically calculated by the radiative transfer model. Trace gases can be acquired from the other operational satellites. When these data are not available in the NRT level products, climatology data could be used in the algorithm.

2.3.1 Primary Sensor Data

1) GOCI-II L1b data

Calibrated/Navigated GOCI-II L1b reflectances, geolocation (latitude/longitude, satellite view angle, satellite azimuth angle, height) information and GOCI-II sensor quality flags are used as the sensor input data for the algorithm.

- pixel location: Level-1B geometry.
- bands: Number 1~12.
- file format: NetCDF-CF convention.
- data type: 32bit float.

2.3.2 Ancillary Data

1) Look-up Table (LUT)

The radiative transfer process in the atmosphere including scattering and absorption due to various atmospheric composition has been estimated for the atmospheric correction. A vector version of radiative transfer model of the Second Simulation of a Satellite Signal in the Solar Spectrum (6SV) (Kotchenova et al., 2006) was used in order to determine characteristics with respect to input data. The simulation tests determined a level of changes in top-of-atmosphere (TOA) reflectance according to input conditions as explained in section 3.4.2.1.

2) Absorbing gases

Absorbing gas concentrations with the spectral absorbing coefficients are required to be estimated the atmospheric transmittance. Current algorithm uses NO₂ and O₃ data from the OMI L2 products. H₂O data from the typical numerical weather prediction model data such as NCEP reanalysis.

2.3.3 Derived Data

GOCI-II surface reflectance algorithm requires light-absorbing aerosol correction. When the L2 aerosol products are available, the aerosol products can be directly used as input data.

2.4 Theoretical Description

Surface reflectance can be calculated from TOA reflectance (R_{TOA}) with regard to GOCI-II wavelengths based on the eq (1).

$$R_{TOA} = \frac{\pi L}{F_0 \cos \theta_0} \quad \text{Eq (1)}$$

Where F_0 refers to extraterrestrial irradiance (E_{sun}), θ_0 is a solar zenith angle (SZA), and L is radiance observed from the sensor. TOA reflectance is affecting the solar radiation through the atmospheric transmission and surface reflection as shown in Figure 2. TOA reflectance calculated in Eq. (1) can be expressed in the following radiative transfer equation (Fraser and Kaufman, 1985).

$$R_{TOA}(\theta_0, \theta_v, \phi) = R_{atm}(\theta_0, \theta_v, \phi) + \frac{TR_{sfc}(\theta_0, \theta_v, \phi)}{1 - R_{sfc}(\theta_0, \theta_v, \phi)S^*} \quad \text{Eq (2)}$$

Where, θ_0, θ_v, ϕ represent a SZA, viewing zenith angle (VZA), and relative azimuth angle (RAA), respectively. R_{atm}, R_{sfc} are atmospheric and surface reflectance. $T(\theta_0), T(\theta_v)$ are atmospheric

transmittance (up and down, respectively), and S^* is spherical albedo of the atmosphere. To estimate the surface reflectance, Eq. (2) can be arranged into Eq (3).

$$R_{sfc}(\theta_0, \theta_v, \phi) = \frac{R_{TOA}(\theta_0, \theta_v, \phi) - R_{atm}(\theta_0, \theta_v, \phi)}{T + S^*(R_{TOA}(\theta_0, \theta_v, \phi) - R_{atm}(\theta_0, \theta_v, \phi))} \quad \text{Eq (3)}$$

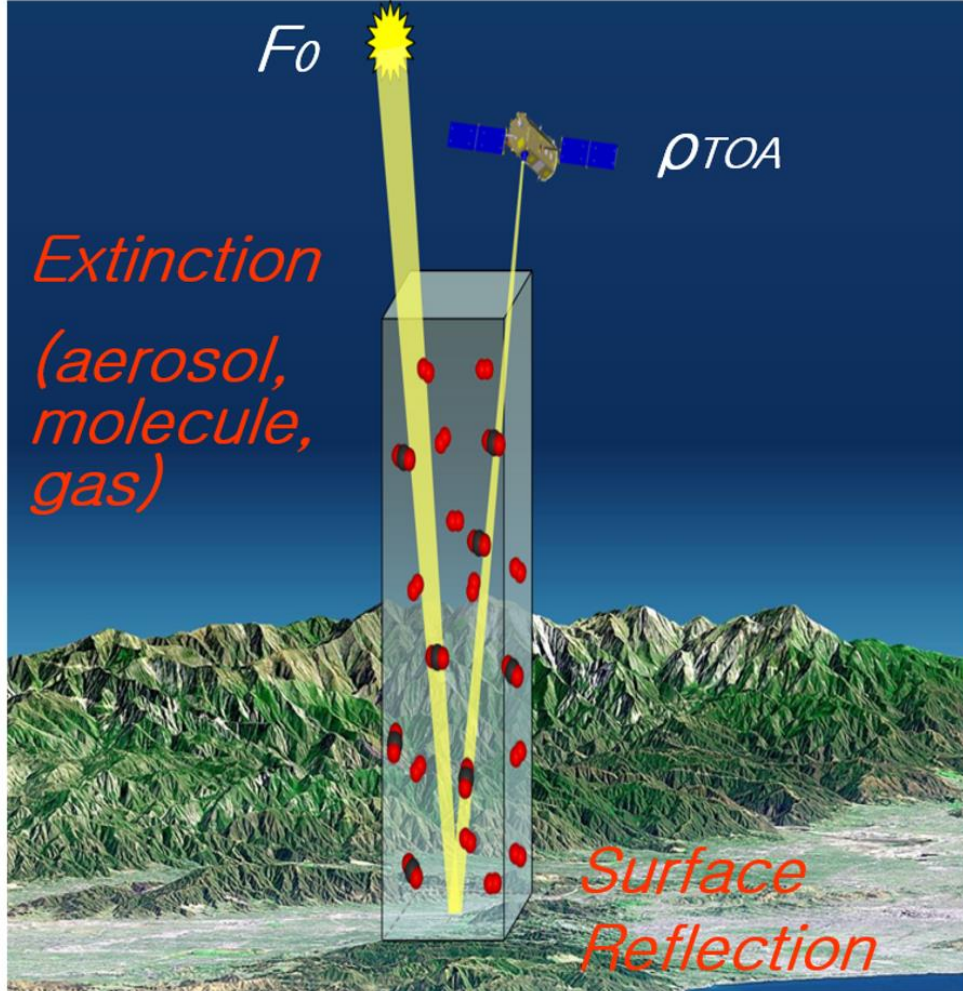


Figure 2. TOA reflectance by radiative transfer process.

2.5 Mathematical Description

2.5.1 Structure and Interpolation of LUT data

In the GOCI-II surface reflectance retrieval algorithm, the vector version 6S radiative transfer code (6SV) (Kotchenova et al., 2006) are used to obtain these variables for the atmospheric correction. In order to produce the LUT, solar zenith angle (9), viewing zenith angle (17), relative azimuth angle (19), and wavelength (12) are considered as input parameters. The effects of air molecule scattering are removed from TOA reflectance based on the LUT interpolated according to a viewing zenith angle.

Table 4. Descriptions on the variables for LUT calculations.

Variable	Data
Wavelength	380, 412, 443, 490, 510, 555, 620, 660, 680, 709, 745, 865 nm
SZA	0°~80°, 10° interval (9 values)
VZA	0°~80°, 5° interval(17 values)
RAA	0°~180°, 10° interval(19 values)
Model Atmosphere	Spring, fall : standard atmosphere summer : midlatitude-summer winter : midlatitude-winter

2.5.2 GOCI-II Surface Reflectance (G2SR)

Surface reflectance can be calculated numerically by correction of atmosphere transmittance from L1b reflectance data. The effect of absorption by trace gases and scattering by air molecules are corrected by using the ancillary datasets. In the GOCI-II surface reflectance retrieval algorithms, the atmospheric correction method and minimum reflectance method using RTMs were applied to minimize the atmospheric transmission issue. Atmospheric correction of satellite measurement is an essential process in the surface reflectance retrieval to remove the effect of gaseous absorption as well as correcting for the effect of an atmospheric molecular and particulate scattering. In order to conduct the atmospheric correction, the Eq. (3) is used with variables such as atmospheric reflectance, atmospheric transmittance, and spherical albedo which could be obtained from satellite measurement and radiative transfer model.

2.5.2.1 Correction of gaseous absorption

Transmittance by trace gases ($\tau_{gas,s}$, $\tau_{gas,v}$) are defined as Eq (4) and (5).

$$\tau_{gas,s} = \exp[-1 \times K_{gas} * Gas * m_s] \quad \text{Eq (4)}$$

$$\tau_{gas,v} = \exp[-1 \times K_{gas} * Gas * m_v] \quad \text{Eq (5)}$$

Table 5. absorption coefficients of trace gases for GOCI wavelengths

Wavelengths [nm]	412	443	490	555	660	680	745	865
kO ₃ [cm ² /mol]	2.24593 7e-23	1.73444 1e-22	7.71921 4e-22	3.46837 8e-21	2.043212 e-21	1.314412 e-21	4.267706 e-22	6.587658 e-23
kNO ₂ [cm ² /mol]	5.38087 e-19	4.12161 e-19	2.84193 e-19	7.05040 e-20	1.03374e- 20	1.12246e- 20	0	0
k _a H ₂ O [m ⁻¹]	0.00455 056	0.00706 914	0.01500 00	0.05960 00	0.410000	0.465000	2.83380	4.60520
k _b H ₂ O [m ⁻¹]	0.00665 000	0.00487 235	0.00316 451	0.00185 907	0.000889 028	0.000783 124	0.000531 599	0.000282 500

*will be updated after the acquisition of the GOCI-II RSR

2.6 Algorithm Product Output

2.6.1 Output

- Surface reflectance at GOCI-II observation time.
- Atmospheric transmittance, QA/QC flag.

2.6.2 Flags

Quality flags for the GOCI-II surface reflectance product are 32bit flag as listed in Table 6.

Table 6. Quality flags in the algorithm output

	Flag name	Flag description
1	Cloud or Ice	Cloud, snow/ice (Masking)
2	Land	Land or water
3	ACR Fail	Algorithm failure
4	L1B outlier	Abnormal reflectance values
5	No O3	Ancillary data(O ₃) not available
6	No NO2	Ancillary data(NO ₂) not available
7	No H2O	Ancillary data(H ₂ O) not available
8	High SZA	Too low sun elevation angle
9	High VZA	Too low satellite elevation angle
10	High AOT	Larger aerosol optical depth
11	Neg. Ref	Negative reflectance
12	-reserved-	
13	-reserved-	
14	-reserved-	
15	-reserved-	
16	-reserved-	
17	-reserved-	
18	-reserved-	
19	-reserved-	
20	-reserved-	
21	-reserved-	
22	-reserved-	
23	-reserved-	
24	-reserved-	

	Flag name	Flag description
25	-reserved-	
26	-reserved-	
27	-reserved-	
28	-reserved-	
29	-reserved-	
30	-reserved-	
31	-reserved-	
32	-reserved-	

2.6.3 Metadata

The following metadata information is included in the output:

- Product Name
- Date, Time
- Effective pixel numbers for the minimum reflectance test
- Image dimension (horizontal pixel * vertical pixel * bands)
- Statistics in image processing
 - Surface reflectance process time
 - Data type

3 Test Data Sets And Output

3.1 Simulated / Proxy Input Datasets

The validation of the surface reflectance products contains the algorithm code implementation and product accuracy test. The former uses simulated input data as truth values. The latter uses measurement data, simulation data, the other operational satellite data. In detail, satellite-retrieved surface reflectance with reference (“truth”) values, and it involves calculation of certain metrics that characterize the level of agreement between the satellite retrievals and reference values. The reference data include (1) surface reflectance data used in the simulation of GOCI-II radiances, and (2) measurements of surface reflectance properties made from the ground measurements. Evaluation is also done by performing retrievals from model-simulated proxy GOCI-II data, and by comparing the retrieved aerosol product to the known surface reflectance input used in the forward simulation.

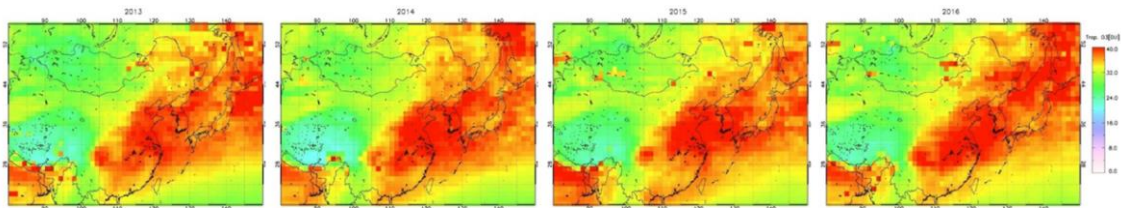


Figure 3. Yearly mean OMI O3 column concentrations (DU) for 2013-2016.

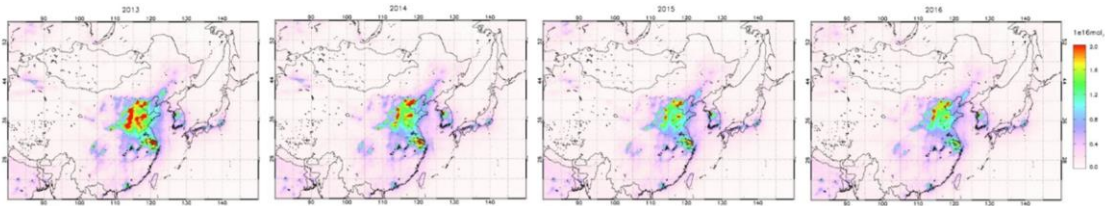


Figure 4. Yearly mean OMI NO2 column concentrations(mol/cm²) for 2013-2016.

3.2 Output from Simulated Datasets

3.2.1 Accuracy Estimates

Performance of the GOCI-II surface reflectance algorithm is measured by precision and accuracy of the retrievals. Accuracy means the mean difference (bias, systematic error) between the retrieved and reference values. Precision is defined as the standard deviation of the differences between GOCI-II and reference data representing the random error of GOCI-II products. In addition, the root mean square error (RMSE) of the differences and correlation between GOCI-II surface reflectance and reference data, are also calculated.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (X_i - X_m)^2}{n}}$$

For the purpose of comparing the surface reflectance obtained with GOCI L1b data, coinciding MODIS level 2 surface reflectance products (code name MOD09/MYD09) were selected. Since acquisition of the ground-truth data is not easy, MODIS L2 products having a known retrieval

accuracy of $\pm(0.005+0.05\text{reflectance})$ (Vermote et al., 2015) are often used to compare the retrieved data. In general, the GOCI surface reflectance is relatively in good agreement with MODIS surface reflectance product as shown in Figure 2. Based on this result, once elevation correction is complete, the final output products will be produced by applying the other GOCI-II surface reflectance retrieval algorithms. For the initial test period, statistical results of RMSE=0.023, BIAS=0.03 are acquired.

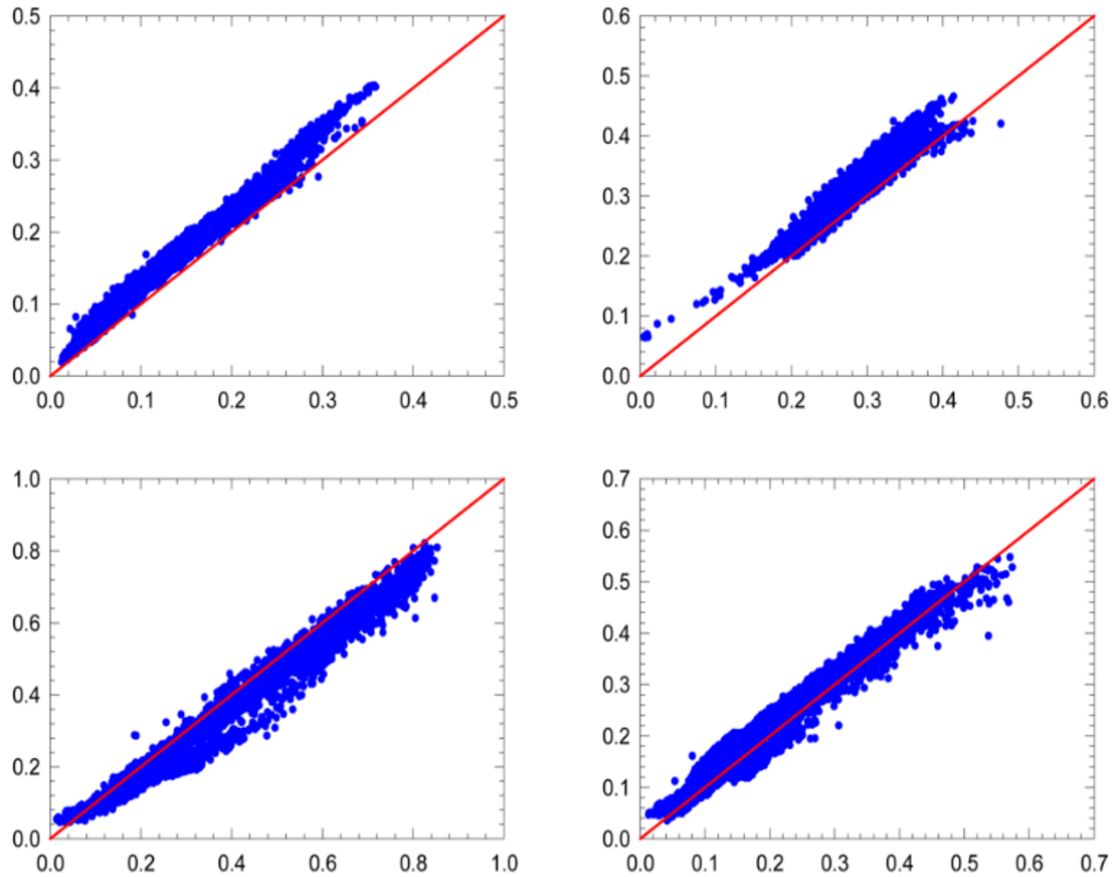


Figure 5. Scatter plots for the GOCI-II surface reflectance products (at 440, 555, 680, 865nm) with MODIS L2 surface reflectance.

3.2.2 Error Budget

Anticipated errors in the GOCI-II surface reflectance algorithm are following.

- Cloud contamination.
- Land cover change during a given period.
- Over/under-estimated atmospheric correction.

4 PRACTICAL CONSIDERATIONS

4.1 Numerical Computation Considerations

1) Algorithm SW process time/memory use optimization

Development environment for the GOCI-II surface reflectance algorithm requires is PC/Workstation with 16GB RAM. Data process time under ADE is less than 10 minutes. NRT process (using the output from the preprocess (~40 min) in the yesterday) can finished within 10 minutes.

4.2 Programming and Procedural Considerations

The original algorithm was developed using the IDL(Version 8.5). For the application to the G2GS ADE, the original version has been ported to C programming version. The surface reflectance retrieval is carried out pixel by pixel. It requires calibrated and geo-located L1b reflectances as dynamic input. The data that need to be configurable for possible post-launch adjustment include algorithm coefficients (LUT), criterion values (e.g. selecting atmospheric conditions), water vapor and ozone dataset depending on the GOCI-II data availability, and metadata setting.

4.3 Quality Assessment and Diagnostics

The following flags will be produced:

- Missing/No data
- Pixel with large local zenith angle (> 60 degrees)
- Less than maximum number of channels used for retrieval
- Large solar zenith angles
- Negative retrievals
- Retrievals over bright surfaces
- Flags based on spatial variability tests for residual cloud contamination

4.4 Exception Handling

- The quality control flags for surface reflectance retrieval will be checked and inherited from the flagged Level 1b sensor input data, including bad sensor input data, missing sensor input data and validity of each GOCI-II channel; and will be checked and inherited from the cloud mask at each pixel.
- Input data sizes are huge for the algorithm, dynamic memory space is required.

4.5 Algorithm Validations

During the pre-launch phase of development, and testing, the primary means of surface reflectance product validation is the comparison with the other operational satellite products such as MODIS L2 products. Because of the limited ground measurements, this type of validation is not available in the present phase. Thus, cross comparison is ideal for evaluating algorithm performance.

5 ASSUMPTIONS AND LIMITATIONS

This section describes the limitations and assumptions in the current version of the surface reflectance retrieval algorithm, and potential future enhancements to the algorithm.

5.1 Performance

The following assumptions have been made in the current algorithm developing:

1. Ancillary data are available for the hourly data processing.
2. Calibrated L1b data available.
3. Assume standard atmosphere for trace gas absorption.
4. Geometry of LUT should be compatible with GOCI-II view range.

5.2 Assumed Sensor Performance

GOCI-II surface reflectance algorithm assumes the sensor will meet its current specifications and produce calibrated quality radiance in the required channels. Currently, the algorithm assumes that calibration uncertainty is less than 0.1% and no difference between measured and calculated values. Also, assumptions in the image acquisition time and spatial differences are used. However, these assumptions are different from the new information, the updated version of algorithm should be carried out.

6 Pre-Planned Product Improvement

- LUT will be updated with obtaining the most recent sensor characteristics.
- When OMI L2 data is not available, other satellite data will be used.

7 REFERENCES

- Chin, M., R. A. Kahn, and S. E. Schwartz (2009), CCSP 2009: atmospheric aerosol properties and climate impacts, Report by the US Climate Change Science Program and the Subcommittee on Global Change Research, National Aeronautics and Space Administration, Washington, DC.
- Dickinson, R. E. (1983), Land surface processes and climate—Surface albedos and energy balance, *Advances in Geophysics*, 25, 305-353.
- Fraser, R. S., and Y. J. Kaufman (1985), Relative importance of aerosol scattering and absorption in remote sensing, *IEEE transactions on geoscience and remote sensing*, 23(5), 625-633.
- Herman, J., and E. A. Celarier (1997), Earth surface reflectivity climatology at 340–380 nm from TOMS data, *Journal of Geophysical Research: Atmospheres*, 102(D23), 28003-28011.
- Kleipool, Q., M. Dobber, J. de Haan, and P. Levelt (2008), Earth surface reflectance climatology from 3 years of OMI data, *Journal of Geophysical Research: Atmospheres*, 113(D18).
- Koelemeijer, R., J. De Haan, and P. Stammes (2003), A database of spectral surface reflectivity in the range 335–772 nm derived from 5.5 years of GOME observations, *Journal of Geophysical Research: Atmospheres*, 108(D2).
- Noguchi, K., A. Richter, V. Rozanov, A. Rozanov, J. Burrows, H. Irie, and K. Kita (2014), Effect of surface BRDF of various land cover types on geostationary observations of tropospheric NO₂, *Atmospheric Measurement Techniques*, 7(10), 3497-3508.
- Veefkind, J. P., J. F. de Haan, E. J. Brinksma, M. Kroon, and P. F. Levelt (2006), Total ozone from the Ozone Monitoring Instrument (OMI) using the DOAS technique, *IEEE transactions on geoscience and remote sensing*, 44(5), 1239-1244.
- Vermote, E. F., N. Z. El Saleous, and C. O. Justice (2002), Atmospheric correction of MODIS data in the visible to middle infrared: first results, *Remote Sensing of Environment*, 83(1), 97-111.
- Vermote, E. F., D. Tanré, J. L. Deuze, M. Herman, and J.-J. Morcette (1997), Second simulation of the satellite signal in the solar spectrum, 6S: An overview, *IEEE transactions on geoscience and remote sensing*, 35(3), 675-686.

8 APPENDIX

End of Document